

UGA Mock-2

Time Allowed: 2 Hours

Maximum Marks: 120

Instructions: Each question has exactly one correct option.

1. Let $P(x)$ be a polynomial of degree 10 such that $P(k) = 2^k$ for $k = 0, 1, 2, \dots, 10$. What is the value of $P(11)$?
(A) 2046 (B) 2047 (C) 2048 (D) 2049
2. Let z_1, z_2, z_3 be complex numbers such that $|z_1| = |z_2| = |z_3| = 1$ and $z_1 + z_2 + z_3 = 0$. Evaluate $|z_1^2 + z_2^2 + z_3^2|$.
(A) 0 (B) 1 (C) 2 (D) 3
3. The number of positive integer pairs (x, y) that satisfy the equation $x^2 - y^2 = 2^{2026}$ is
(A) 1011 (B) 1012 (C) 1013 (D) 2025
4. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function satisfying $f(x+y) = f(x) + f(y) + xy$ for all $x, y \in \mathbb{R}$, and $f(1) = 1$. Find the value of $f(-1)$.
(A) -1 (B) 0 (C) 1 (D) $1/2$
5. Evaluate
$$\int_{-\pi/4}^{\pi/4} \frac{x^7 - x^5 + x^3 - x + 1}{\cos^2 x} dx.$$

(A) 1 (B) 2 (C) $\pi/2$ (D) $\pi/4$
6. Let A be a 3×3 matrix with real entries such that $A^3 = -I$. Which of the following cannot be the trace of A ?
(A) -1 (B) 1 (C) 2 (D) -2
7. The number of subsets of the set $\{1, 2, \dots, 10\}$ that contain no two consecutive integers (including the empty set) is
(A) 233 (B) 89 (C) 55 (D) 144

8. The highest power of 3 that divides $(3^n)!$ is
- (A) $\frac{3^n-1}{2}$ (B) $\frac{3^{n+1}-1}{2}$ (C) 3^{n-1} (D) $\frac{3^n-1}{2} - n$
9. Let ABC be a triangle with side lengths a, b, c such that a^2, b^2, c^2 are in arithmetic progression. Then $\cot A, \cot B, \cot C$ are in
- (A) A.P. (B) G.P. (C) H.P. (D) None
10. If $\sum_{i=1}^n \cos^{-1}(x_i) = 0$, what is the value of $\sum_{i=1}^n x_i$?
- (A) $n/2$ (B) $-n$ (C) 0 (D) n
11. Let $z_1, z_2, \dots, z_n$ be the n distinct complex roots of the equation $z^n - 1 = 0$. Evaluate $\sum_{1 \leq i < j \leq n} |z_i - z_j|^2$.
- (A) $2n^2$ (B) $n(n-1)$ (C) $n^2/2$ (D) n^2
12. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuously differentiable function such that $f(0) = 0$ and $f(1) = 1$. What is the minimum possible value of the integral $\int_0^1 (f'(x))^2 (1+x^2) dx$?
- (A) $\pi/4$ (B) $4/\pi$ (C) $2/\pi$ (D) $\pi/2$
13. Let S be the set of all integers n such that $n^2 + 20n + 11$ is a perfect square. The number of elements in S is
- (A) 2 (B) 4 (C) 6 (D) 8
14. How many subsets of the set $S = \{1, 2, \dots, 2000\}$ have the property that the sum of their elements is divisible by 5? (Assume the sum of the elements of the empty set is 0).
- (A) $\frac{1}{5}(2^{2000} + 2^{402})$ (C) $\frac{1}{5}(2^{2000} + 2^{400})$
 (B) $\frac{1}{5}(2^{2000} - 2^{402})$ (D) $\frac{1}{5}(2^{2000} - 2^{400})$
15. Let $x_1, x_2, \dots, x_{100}$ be the complex roots of the polynomial equation $x^{100} + x^{99} + \dots + x + 1 = 0$. The value of $\sum_{i=1}^{100} x_i^{2026}$ is
- (A) 0 (B) 100 (C) -1 (D) 1
16. The number of non-negative integer solutions to the equation $x + y + z + w = 10$ is
- (A) 286 (B) 1001 (C) 715 (D) 84
17. Let $g(x) = \int_0^x (t-1)(t-2)^2 dt$. The function $g(x)$ has a local minimum at
- (A) $x = 1$ (B) $x = 2$ (C) $x = 1$ and $x = 2$ (D) Nowhere

18. Consider a regular n -gon. It is given that exactly 320 distinct triangles can be formed whose vertices are vertices of the n -gon and which share exactly one side with the original n -gon. Find the value of n .

(A) 16 (B) 18 (C) 20 (D) 22

19. Evaluate the following

$$\lim_{n \rightarrow \infty} \left(\frac{(2n)!}{n! \cdot n^n} \right)^{\frac{1}{n}}$$

(A) $4/e$ (B) $e/4$ (C) $2/e$ (D) $e/2$

20. Let $p > 3$ be a prime number. The sum $1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{p-1}$ is expressed as a fraction $\frac{a}{b}$ in lowest terms. What is the remainder when the numerator a is divided by p^2 ?

(A) 0 (B) 1 (C) p (D) $p - 1$

21. Let $f : \mathbb{N} \rightarrow \mathbb{N}$ be a strictly increasing function satisfying the functional equation $f(f(n)) = 3n$ for all $n \in \mathbb{N}$. Find the value of $f(2026)$.

(A) 3891 (B) 3981 (C) 2755 (D) 4052

22. Let S be the set of all complex roots of the equation $z^{24} - 1 = 0$. How many distinct equilateral triangles can be formed whose vertices all belong to S ?

(A) 4 (B) 6 (C) 8 (D) 12

23. In $\triangle ABC$, let $AB = c$, $BC = a$, and $CA = b$. If $\angle A = 3\angle B$, which of the following relations must hold true?

(A) $a^3 = b^3 + c^3$ (C) $a^2c = (b - c)^2(b + c)$
 (B) $a^2 = b(b + c)$ (D) $bc^2 = (a - b)(a + b)^2$

24. What is the remainder when the polynomial x^{2026} is divided by $x^2 - x + 1$?

(A) $x - 1$ (B) $-x$ (C) $-x + 1$ (D) 1

25. Let D_k denote the number of derangements of k objects. The number of permutations of the set $\{1, 2, \dots, n\}$ that possess exactly one fixed point is

(A) $n \cdot D_{n-1}$ (B) $D_n - 1$ (C) $n! - D_n$ (D) $\frac{n!}{e}$

26. Evaluate

$$\lim_{n \rightarrow \infty} n^2 \int_0^{1/n} x^{x+1} dx$$

(A) $1/2$ (B) $1/3$ (C) 0 (D) 1

- 27.** A fair coin is tossed repeatedly. What is the expected number of tosses required to observe the sequence "Heads-Tails-Heads" (HTH) for the very first time?
- (A) 8 (B) 10 (C) 12 (D) 14
- 28.** The largest prime factor of the number $3^{12} + 4$ is
- (A) 73 (B) 193 (C) 677 (D) 37
- 29.** Let M be a 3×3 real matrix satisfying $M^T M = I$ (orthogonal) and $\det(M) = 1$. What is the value of $\det(M - I)$?
- (A) 2 (B) 1 (C) -1 (D) 0
- 30.** Let $S = \{x \in \mathbb{R} : \sin(\sin x) = x\}$. The total number of elements in the set S is
- (A) 1 (B) 3 (C) Infinite (D) 0

Answer Key

1. A	7. D	13. B	19. A	25. A
2. A	8. A	14. C	20. A	26. A
3. B	9. A	15. C	21. B	27. B
4. B	10. D	16. A	22. C	28. C
5. B	11. D	17. A	23. C	29. D
6. C	12. B	18. C	24. B	30. A